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1)

$$2\,NaHCO_3(aq) \xrightarrow{varme} Na_2CO_3(aq) + H_2O(g) + CO_2(g)$$

fordi reaksjonen skjer i et åpent kar vil gassen diffundere ut i luften derfor vil ikke likevekt kunne innstille seg.

2) i) is er et fast stoff → 1

ii) $Cl_2(g)$ $\quad a_{Cl_2} = P_{Cl_2} / P_{ref} = 1\,atm / 1\,bar = 1{,}013 \cdot 10^5 / 10^5 = 1{,}013$

iii) $a_{CH_3OH} = c_{CH_3OH} / c_{ref} \rightarrow a_{CH_3OH} = 0{,}1\,mol/l / 1\,mol/l = 0{,}1\,mol/l$

3) $k_p = 64$

hvis $Q < k$ → høyre
-"- $Q > k$ → venstre

$$BrF_3(g) \rightleftharpoons BrF(g) + F_2(g)$$

$p = 0{,}52\,atm \quad p = 5{,}71\,atm \quad p = 1{,}36\,atm$

$$Q = \frac{P_{BrF} \cdot P_{F_2}}{P_{BrF_3}} = \frac{(5{,}71 \cdot 1{,}36)\,atm^2}{0{,}52\,atm} = 14{,}9\,atm$$

$Q < K_p$ ⇒ netto reaksjonsretning er høyre

4) $k = 0{,}143$ $\quad \frac{[\text{metylsykloheksan}]}{[\text{sykloheksan}]}$ $\quad (0{,}1+x)/(0{,}2-x) = 0{,}143 \quad | \cdot (0{,}2-x)$

$0{,}1 + x = 0{,}143(0{,}2-x)$

$0{,}1 + x = 0{,}0286 - 0{,}143x$

$1{,}143x = 0{,}0286 - 0{,}1 = -0{,}0714 \quad | \div 1{,}143$

$x = -0{,}0625$

	Start	likevekt
Sykloheksan	0,2 mol/l	$0{,}2 - x$
metyl	0,1 mol/l	$0{,}1 + x$

finner likevektskonsentrasjonen av sykloheksan

$0{,}2 - x \rightarrow 0{,}2 - (-0{,}0625) = 0{,}263\,mol/l$

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5) a) $NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$ $\quad K_p = \frac{P_{NH_3} \cdot P_{HCl}}{P_{NH_4Cl}}$

$P_{tot} = 4,4$ atm

totaltrykket er 4,4 atm, siden det er et lukket system og forholdet mellom NH_3 & HCl er 1:1, så antar jeg at $P_{NH_3} = P_{HCl} = 2,2$ atm

$K_p = \frac{2,2\,atm \cdot 2,2\,atm}{1} = 4,84\,atm^2 = \underline{\underline{4,8\,atm^2}}$

fordi NH_4Cl er salt

b) $P_{tot} = 4,4\,atm \cdot 1,013 = 4,4572$ bar $\rightarrow P_{NH_3} = P_{HCl} = 2,2286$ bar

$K_p = \frac{(2,2286\,bar)^2}{1} = 4,9667 = \underline{\underline{5,0\,bar^2}}$

c) $k = \frac{a_{NH_3} \cdot a_{HCl}}{a_{NH_4Cl}}$ $\quad a_{NH_3} = a_{HCl} = \frac{2,2\,atm \cdot 1,013 \cdot 10^5}{10^5\,Pa} = 2,286$

$k = \frac{(2,2286)^2\,atm^2}{1} = \underline{\underline{5,0\,atm^2}}$

6) ~~T = 375 K~~ ~~$K_p = 2,4$~~ ~~$V = 1\,dm^3$~~ ~~P_{tot}?~~ ~~$pV = nRT$~~

$SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$

$m = 6,7\,g$

$M_m = 134,97\,g/mol$

$n = m/M_m = 0,0496\,mol$

$r = 15,261$ atm

a) $P_{SO_2Cl_2} = \frac{0,0496\,mol \cdot 8,314\,Nm\,mol^{-1}K^{-1} \cdot 375\,K}{1 \cdot 10^{-3}\,m^3} = 1546400,4\,N/m^2$

ved likevekt er $n_{SO_2Cl_2} = 0,0248$ og $n_{SO_2} = n_{Cl_2} = 0,0124$

$p = \frac{nRT}{V} \rightarrow P_{SO_2Cl_2} = \frac{0,0248\,mol \cdot 8,314\,Nm\,mol^{-1}K^{-1} \cdot 375\,K}{1 \cdot 10^{-3}\,m^3} = 77320,2\,Pa = 0,763\,atm$

$P_{SO_2Cl_2} = \frac{0,0124\,mol \cdot 8,314\,Nm\,mol^{-1}K^{-1} \cdot 375\,K}{1 \cdot 10^{-3}\,m^3} = 38660,1\,Pa = 0,3815\,atm$

$P_{tot} = (0,763 + 2 \cdot 0,382)\,atm =$

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6) a) $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$

$m = 6{,}7\,g$

$M_m = 134{,}97\,g/mol$

$n = \frac{m}{M_m} = 0{,}0496\,mol$

$k_p = 2{,}4$ $\quad k = 375\,K$ $\quad V = 1\,dm^3$ $\quad pV = nRT$

$p_{SO_2Cl_2} = ?$ $\quad p_{SO_2} = ?$ $\quad p_{Cl_2} = ?$

$$p_{SO_2Cl_2} = \frac{0{,}050\,mol \cdot 8{,}314\,Nm\,mol^{-1}K^{-1} \cdot 375\,K}{1 \cdot 10^{-3}\,m^3} = 155887{,}5\,Pa = \underline{\underline{1{,}5385\,atm}}$$

$p_{SO_2Cl_2} = 1{,}5385\,atm$ med start

ved likevekt:

$$\frac{(0+x)(0+x)}{1{,}54-x} = 2{,}4 \rightarrow x^2 = 2{,}4(1{,}54-x)$$

$$x^2 + 2{,}4x = 3{,}696 \rightarrow x^2 + 2{,}4x - 3{,}70$$

$$x = \frac{(-2{,}4) \pm \sqrt{2{,}4^2 - 4(-3{,}70)}}{2} = \frac{-2{,}4 \pm 4{,}53}{2}$$

$x_- = -3{,}465$ $\quad x_+ = 1{,}065$

↑ reaksjonen gir mat høyre, så denne verdien kan ikke brukes

$p_{SO_2} = p_{Cl_2} = 1{,}07\,atm$ $\quad p_{SO_2Cl_2} = (1{,}53 - 1{,}07)\,atm = 0{,}46\,atm$

$p_{tot} = (1{,}07 \cdot 2 + 0{,}46)\,atm = \underline{\underline{2{,}6\,atm}}$

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6) h) $K_c = \frac{[SO_2]\cdot[Cl_2]}{[SO_2Cl_2]} = \frac{(n_{SO_2}/V)\cdot(n_{Cl_2}/V)}{(n_{SO_2Cl_2}/V)} = \frac{n_{SO_2}\cdot n_{Cl_2}}{n_{SOCl_2}}\cdot V$

$n = PV/RT$ $\quad P_{SO_2Cl_2} = 0{,}46\,atm = 46598\,Pa$

$P_{SO_2} = P_{Cl_2} = 1{,}07\,atm = 108391\,Pa$

$n_{SO_2Cl_2} = \frac{46598\,N/m^2 \cdot 1\cdot10^{-3}\,m^3}{8{,}314\,Nm\,mol^{-1}K^{-1}\cdot 375\,K} = 0{,}0149 \approx 0{,}015\,mol$

$n_{SO_2} = n_{Cl_2} = \frac{108391\,N/m^2 \cdot 1\cdot10^{-3}\,m^3}{8{,}314\,Nm\,mol^{-1}K^{-1}\,375\,K} = 0{,}0348 \approx 0{,}035\,mol$

$C_{SO_2Cl_2} = \frac{n}{V} = \frac{0{,}015\,mol}{1\,L} = 0{,}015\,mol/L$

$C_{SO_2} = C_{Cl_2} = \frac{0{,}035\,mol}{1\,L} = 0{,}035\,mol/L$

$K_c = \frac{(0{,}035)^2}{0{,}015} = \underline{\underline{0{,}082\,mol/L}}$ ← avrundningsfeil?